

Liability, Verification, and the Market for Expertise in the Presence of AI

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Abstract

Artificial intelligence makes competent execution cheap, so for tasks a principal can verify himself he increasingly performs them directly rather than hiring an intermediary. For tasks he cannot verify, such as a diagnosis or an audit, hiring remains necessary. But what made hiring valuable, the ability to tell genuine expertise from imitation, is exactly what a commoditized AI erodes: once anyone can produce a fluent-looking deliverable, fluency stops being informative. A liability bond can restore separation between genuine expertise and imitation, but only if the bond is enforceable, and enforceability runs into the same verification problem that created the need for the signal. We build a model that nests a make-or-buy decision inside a liability-bond signaling game. Separation occurs only in an intermediate band of verifiability. When verification is high, the principal does the task himself. When it is low, even a known imitator, posting a large enough bond, is preferable to doing the task himself, so no bond can screen it out. A limit on the capital a genuine expert can post raises the band's lower edge.

Keywords: liability, signaling, credence goods, verification, artificial intelligence, delegation, judgment-proof problem

JEL codes: D82, D86, K13, L84, O33

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1 Introduction

Should a firm hire a consultant to run an analysis, or run it itself with an AI model? The answer used to turn mostly on capability. It increasingly turns on whether the firm can *tell*, once the work is done, whether it is any good. For a webpage, a small software feature, or a first-pass slide deck, the answer is close to yes: the principal can click through it or compare it against a brief. For a tax opinion on a novel transaction, an audit of a firm’s financial statements, or a medical diagnosis, the answer is closer to no, because judging the output requires much the same expertise that produced it. The first class of tasks is drifting toward direct AI use. Hiring survives in the second, where a genuine problem is opening up.

The problem is Akerlof’s market for lemons, restaged: a principal who cannot verify quality must rely on some correlate he *can* observe, and the market unravels unless a credible signal separates types. What is new is that AI dissolves the correlates. A polished deliverable or a fluent report used to signal expertise because it was costly to produce without it, and a capable AI makes it cheap for anyone. At the same time, verification is getting harder, because AI-assisted outputs are sophisticated enough that only genuine expertise can check them.

Liability is a classical response: make the seller bear the cost of failure after the fact. A performance bond, forfeited on a provable failure, lets a genuine expert signal credibly even when the principal cannot inspect the work. But the response inherits the disease it is meant to cure. A court’s ability to establish that an audit or a diagnosis was wrong is itself a verification problem, and if the principal cannot tell good work from bad, a court will typically find it hard too. The model captures this shared difficulty in a single parameter, v , the probability that a bad outcome is verified, whether by the principal or, indirectly, by a court enforcing a bond. The question is how liability’s ability to restore separation depends on v , on the value of the task, and on the capital a genuine expert can put behind the signal.

Section 2 sets out a two-stage model. The principal chooses between doing the task himself with an AI and delegating to an intermediary of privately known type, screened through a bond whose enforcement depends on the same v . Section 3 solves the baseline, without a limit on the expert’s capital. Its main result is that separation occurs only in an intermediate band of verifiability. Above the band, the principal does the task himself. Below it, even a known wrapper, posting a large enough bond, beats doing the task himself, so no bond can screen it out. Section 4 adds a capital constraint on bonding, which raises the band’s lower edge. Section 5 discusses implications, Section 6 extensions, and Section 7 concludes.

Related literature. The analytical core combines a [Spence \(1973\)](#) and [Riley \(1979\)](#) signaling game with the [Cho and Kreps \(1987\)](#) intuitive criterion, applied to a liability bond rather than to education. The wealth constraint on bonding in Section 4 is the judgment-proof problem ([Shavell, 1986](#)) applied to the credibility of a signal rather than to ex post compensation. Two recent papers share this paper’s concern with AI, verification, and liability but locate the friction at a different stage. [Gans \(2026\)](#) studies staged access and liability for dual-use AI as a regulatory design problem

facing a defender and an adversary, where the friction is release timing rather than delegation. [Bryan and Gans \(2026\)](#) study how an AI model should be trained given how humans will use it, a friction upstream of the market-structure question asked here. The friction studied here is whether a liability signal can survive once the technology that created the need for it also erodes a court’s ability to enforce it. It is closest in spirit to the credence-goods literature following [Akerlof \(1970\)](#) and [Darby and Karni \(1973\)](#), applied to a setting in which AI makes quality both cheap to fake and hard to verify.

2 Model

2.1 Setup and timing

A risk-neutral principal faces a task worth S if delivered good and $S - L$ if delivered bad, where $L > 0$ is the stake destroyed by a bad outcome. Two objects are drawn before either party acts:

- the task’s *verification level* $v \in [0, 1]$, the probability that inspection, if performed, reveals the true state. This is the same primitive whether inspection is carried out by the principal (DIY) or, indirectly, by a court asked to enforce a bond (delegation);
- the intermediary’s *type* $\theta \in \{H, L\}$, where H is a genuine expert and L is a *wrapper* reselling standard AI access. The type is private to the intermediary; the common prior is $\mu = \Pr(H)$.

Timing.

0. Nature draws v (common knowledge, Assumption 2) and θ (private to the intermediary, prior μ).
1. The principal chooses **DIY**, operating the AI himself and obtaining a good outcome with probability p_0 , or enters the **delegation market**.
2. If he delegates, the matched intermediary posts a bond $B(\theta) \geq 0$, subject to the capacity bound of Assumption 4. The principal observes B but not θ , forms a belief $\mu(B)$, and chooses to **engage** the intermediary at the exogenous price t or to **walk away**, falling back to DIY.
3. If engaged, the outcome is good with probability $p_\theta \equiv 1 - f_\theta$. If it is bad (probability f_θ), it is verified with probability $e(v)$, and B is forfeited to the principal; otherwise the loss goes uncompensated.

DIY’s payoff, which incorporates the verify-and-retry mechanism of Assumption 3 below, is

$$U_{\text{DIY}}(v) = S - (1 - v)f_0L - c_0 - vf_0\kappa, \quad (1)$$

where $f_0 \equiv 1 - p_0$ is the principal’s own (non-expert) failure probability, c_0 his cost of operating the AI directly, and κ the cost of a successful retry when inspection catches a bad draw. If the principal

engages an intermediary of type θ that has posted bond B , expected payoffs are

$$\pi(\theta, B) = t - f_\theta e(v) B \quad (\text{intermediary}), \quad U(\theta, B) = S - t - f_\theta L + f_\theta e(v) B \quad (\text{principal}). \quad (2)$$

An intermediary that is not engaged earns zero. Write $\bar{U}$ for the principal's outside option. Section 3.2 treats $\bar{U}$ as a fixed number; from Section 3.3 on, $\bar{U} = U_{\text{DIY}}(v)$. Types are ordered by $f_L > f_H > 0$, and throughout we maintain $f_L L > t$: a wrapper's expected loss exceeds the fee. The principal engages when indifferent.

2.2 Assumptions

Assumption 1 (Fixed price). *The price t is exogenous and common to both intermediary types.*

Type affects the probability of success, not the price of operating the AI, consistent with the premise that raw AI execution is itself commoditized.

Assumption 2 (Common knowledge of verifiability). *The verification level v is known to both parties before the bond is posted.*

Assumption 3 (Mode-specific remedy). *A caught bad draw is remedied differently depending on who operates the AI. Under DIY, the principal redoes the work at cost κ . Under delegation, the principal cannot redo the intermediary's work; his remedy for a caught bad draw is the bond.*

Retry is the principal's self-correction technology; the bond is delegation's remedy. The distinction reflects what each party brings: the principal can redo his own work when he catches a failure, while what an intermediary offers is a higher probability of getting it right the first time.

Assumption 4 (Capacity bound). *The genuine type cannot credibly post a bond larger than $\bar{w}$.*

This is the judgment-proof problem applied to the credibility of the signal itself, not merely to ex post compensation. Only the genuine type's capacity is bounded; we adopt this asymmetry for simplicity, since the results below concern the genuine expert's ability to post a bond. Assumption 4 is set aside in Section 3 and imposed in Section 4.

Assumption 5 (Equilibrium selection). *Among perfect Bayesian equilibria that survive the intuitive criterion, we focus on (i) pooling without a bond when it satisfies the principal's participation constraint, and otherwise (ii) the least-cost separating equilibrium, when it exists. Where neither exists, separation fails, and we do not characterize the outcome.*

Both rules are selections, not refinement results. Pooling at a positive bond can coexist with separation for some priors; we set it aside, since it asks the genuine type to pay for a bond that does not distinguish it from the wrapper.

The enforcement technology is taken in its simplest form, $e(v) = v$. This is the most direct reading of "if it is hard for the principal to verify, it is hard for a court to verify too" as one primitive rather than a correlated pair. A general affine specification, $e(v) = e_0 + (1 - e_0)v$, is examined in Appendix A.7 and changes only the capacity threshold of Section 4.

3 Baseline results

This section derives the baseline without the capacity bound of Assumption 4. Even when a genuine expert can post whatever bond separation requires, separation is possible only in an intermediate band of verifiability. Section 4 then asks how a limit on that bond changes the band.

3.1 When does the principal do it himself?

Before introducing type uncertainty and a liability bond, it is useful to isolate the paper's first intuition. Suppose for now that the intermediary is of a single, known type that succeeds with probability p_1 , to be compared with the principal's own success probability p_0 . For a given verification level v , does the principal prefer to delegate to this known type at the posted price, or to do the task himself?

Proposition 1. *Suppose there is no uncertainty about the delegate's type (its success probability p_1 is known) and $L > \kappa$. There is a unique threshold*

$$v^* = \frac{\Delta p \cdot L - \Delta c}{f_0(L - \kappa)}, \quad \Delta p \equiv p_1 - p_0, \quad \Delta c \equiv t - c_0, \quad (3)$$

below which delegating at the posted price makes the principal strictly better off than DIY, and above which delegation cannot make him better off than DIY.

Proof. Appendix A.1. □

High verifiability makes catch-and-retry a substitute for expertise: if a bad draw can be cheaply caught and redone, the principal does not need someone who gets it right the first time. Low verifiability makes the retry option worthless, because the principal cannot tell good from bad. The comparison then collapses to whether the quality premium is worth the price premium, which is exactly the terrain on which a credence-good unraveling problem can bite. The threshold is not yet about liability. It establishes that verification technology, not production capability, is what should predict which tasks migrate to direct AI use.

3.2 The liability bond as a signal

Now introduce type uncertainty and set the DIY question aside for the moment. The principal faces an intermediary of unknown type θ , screened through a bond, at a fixed enforcement probability e and a fixed outside option $\bar{U}$.

Proposition 2. *Suppose a known wrapper is not worth engaging even at a zero fee, $S - f_L L \leq \bar{U}$. Then the least-cost bond that separates H from L is*

$$B_H^* = \frac{t}{f_L e}, \quad (4)$$

and H is always willing to post it. It sustains engagement whenever the principal prefers an identified H at this bond to his outside option.

Proof sketch. The marginal cost of posting bond is $f_\theta e$, strictly lower for H since $f_L > f_H$. Single crossing is therefore derived from one type-specific primitive, each type's own probability of a bad draw (f_θ), rather than assumed. A wrapper that reveals itself can obtain engagement only by posting a bond that leaves the principal at least $\bar{U}$, which leaves the wrapper at most $S - f_L L - \bar{U} \leq 0$; its payoff when revealed is therefore zero. Mimicking H's bond is thus deterred exactly when $t - f_L e B \leq 0$, and the least such bond is B_H^* . H's own payoff, $t - f_H e B_H^* = t(1 - f_H/f_L)$, is positive. Full derivation in Appendix A.2. $\square$

Proposition 3. *At B_H^* , the principal's payoff does not depend on e : $\partial U(H, B_H^*)/\partial e = 0$, provided the enforcement technology is costless to run. Any real, additive enforcement or bond-carrying cost breaks this invariance.*

Proof. One substitution: $U(H, B_H^*) = S - t - f_H L + f_H e \cdot t/(f_L e) = S - t - f_H L + f_H t/f_L$, in which e cancels between the grossed-up bond and the recovery term. $\square$

The result is easy to over-read, so it is worth stating precisely. Worse enforcement inflates the *nominal* bond needed to keep deterring the wrapper ($\partial B_H^*/\partial e < 0$), and that inflation exactly offsets the lower probability of collecting it. Expected recovery, and with it the principal's payoff, is untouched. Better courts, in this model, do not make separation more valuable once it is achieved; they make it more *feasible*, provided it is achievable at all — a conclusion that assumes costless bonding, revisited under more realistic costly bonds in Section 6. That qualifier, whether the bond can actually be posted, is what the capacity bound of Assumption 4 puts back into the model, and it is where Section 4 picks up.

Proposition 4. *Pooling at $B = 0$ satisfies the principal's participation constraint if and only if $\mu \geq \mu^*$, where*

$$\mu^* = \frac{f_L - \frac{S-t-\bar{U}}{L}}{f_L - f_H}. \quad (5)$$

Proof. Appendix A.3. $\square$

Is a signal needed at all, or does the pooled belief already satisfy the principal's participation constraint without costly bonding? This is a genuine gate, separate from whether a separating bond, once needed, is affordable. Above μ^* , H has no reason to signal: engaged at $B = 0$, it already earns t , and any bond only lowers that to $t - f_H e B$. This follows from Assumption 1. Because the price is fixed and common to both types, being identified as H is worth nothing beyond being engaged, unlike in Spence (1973), where the wage itself responds to beliefs.

3.3 Nesting the two: the separating band

Substitute $\bar{U} = U_{\text{DIY}}(v)$ from (1) and $e(v) = v$ into Propositions 2–4. This is the paper’s basic interaction: a delegation market screened by a bond, evaluated against an outside option that itself depends on the same verification parameter. Write

$$\Delta \equiv Lf_0 + c_0 - Lf_L \quad (6)$$

for the principal’s expected shortfall, at $v = 0$, relative to a known wrapper working for free. The scope condition of Proposition 2, $S - f_L L \leq U_{\text{DIY}}(v)$, then reads $v \geq v_W \equiv \Delta/[f_0(L - \kappa)]$. Below v_W , the principal would rather engage even a known wrapper, provided it posts a large enough bond, than do the task himself.

Proposition 5. *Suppose $L > \kappa$. The pooling threshold and the participation boundary for an identified genuine expert are*

$$\mu^*(v) = \frac{f_L - f_0 + \frac{t-c_0}{L} + v \frac{f_0(L-\kappa)}{L}}{f_L - f_H}, \quad v_{\text{IR}} = \frac{f_0 L + c_0 - t - f_H L + f_H t / f_L}{f_0(L - \kappa)}, \quad (7)$$

with $\mu^*(v)$ affine and increasing in v , and $v_W < v_{\text{IR}}$. The (v, μ) plane divides into four regions:

- (a) Pooling without a bond: $\mu \geq \mu^*(v)$.
- (b) Separating band: $\mu < \mu^*(v)$ and $v_W \leq v \leq v_{\text{IR}}$. H separates at B_H^* .
- (c) No separation: $\mu < \mu^*(v)$ and $v < v_W$. The wrapper cannot be pushed to a zero payoff, so no bond deters it.
- (d) DIY: $v > v_{\text{IR}}$. Delegation cannot make the principal better off than DIY.

Proof. Appendix A.4. □

The separating band is the paper’s organizing object, and each of its edges has a distinct source. The upper edge is Proposition 1’s threshold reappearing once type uncertainty and a bond are introduced, and the correspondence is exact:

$$v_{\text{IR}} = v^*|_{f_1=f_H} + \frac{f_H t / f_L}{f_0(L - \kappa)}. \quad (8)$$

That is, v_{IR} is the make-or-buy threshold evaluated at the genuine type, shifted up by the expected bond recovery $f_H e B_H^* = f_H t / f_L$ (the e -invariant object of Proposition 3) divided by the slope of U_{DIY} in v . In short, v_{IR} is Proposition 1 plus the bond. Above it, a principal who can catch and redo bad work has no need to identify an expert.¹

¹Past v_{IR} , an identified H can still obtain engagement by posting a bond larger than B_H^* that leaves the principal exactly at $U_{\text{DIY}}(v)$. The bond then works as an expected price rebate, and delegation cannot make the principal better off than DIY.

The lower edge, v_W , is where imitation becomes too good to screen out. Below it, an identified wrapper is still engaged, at a bond of its own, and earns a positive payoff, so no bond can push it to the zero payoff that separation requires. This edge exists without any limit on the genuine expert's capital: even an expert who can post any bond cannot separate from a wrapper that the principal would engage anyway.

Finally, the pooling region shows that a signal is not always *needed*: if the prior is favorable enough, the principal engages at $B = 0$.

4 Capital-constrained bonding

We now impose Assumption 4. The question is how a limit on the bond a genuine expert can post changes the separating band of Proposition 5.

Proposition 6. *Separation at B_H^* is feasible only if $B_H^* \leq \bar{w}$, that is, $v \geq v_{\text{cap}} \equiv t/(f_L \bar{w})$. Under $e(v) = v$, the separating band becomes*

$$\mu < \mu^*(v), \quad \max\{v_W, v_{\text{cap}}\} \leq v \leq v_{\text{IR}},$$

and is empty if $v_{\text{cap}} > v_{\text{IR}}$. For $\mu < \mu^(v)$ and $v < \max\{v_W, v_{\text{cap}}\}$ there is no separation. The pooling and DIY regions of Proposition 5 are unchanged.*

Proof. Appendix A.5. □

As v falls, the bond needed to separate rises (Proposition 2) while the DIY fallback deteriorates (Proposition 1). The capacity bound acts only on the band's lower edge, which is now set by whichever binds first: a wrapper that can be engaged on its own (v_W), or a separating bond the expert cannot afford (v_{cap}). When $v_{\text{cap}} \leq v_W$, capital is slack everywhere it matters and the band is the baseline one. When $v_{\text{cap}} > v_W$, capital rather than imitation sets the edge, and separation is lost on $[v_W, v_{\text{cap}})$, where a genuine expert who could post any bond would be identified. That strip is hatched in Figure 1. It is exactly where identifying the expert is still valuable to the principal ($v \leq v_{\text{IR}}$) and the expert cannot credibly signal itself.

5 Discussion

Four intuitions motivated this paper and were, at the outset, in tension without a resolution mechanism. Genuine expertise commands a premium. That premium is hard to verify once AI execution is a commodity. Liability can restore the signal, since a bond substitutes for direct verification. But enforcement of that liability is hostage to the same verification problem the bond exists to route around. The model shows *when* these forces resolve in favor of separation rather than merely asserting that they are in tension. The answer is a band. Verification must be low enough that the principal still needs an expert, high enough that a wrapper cannot be engaged on its own,

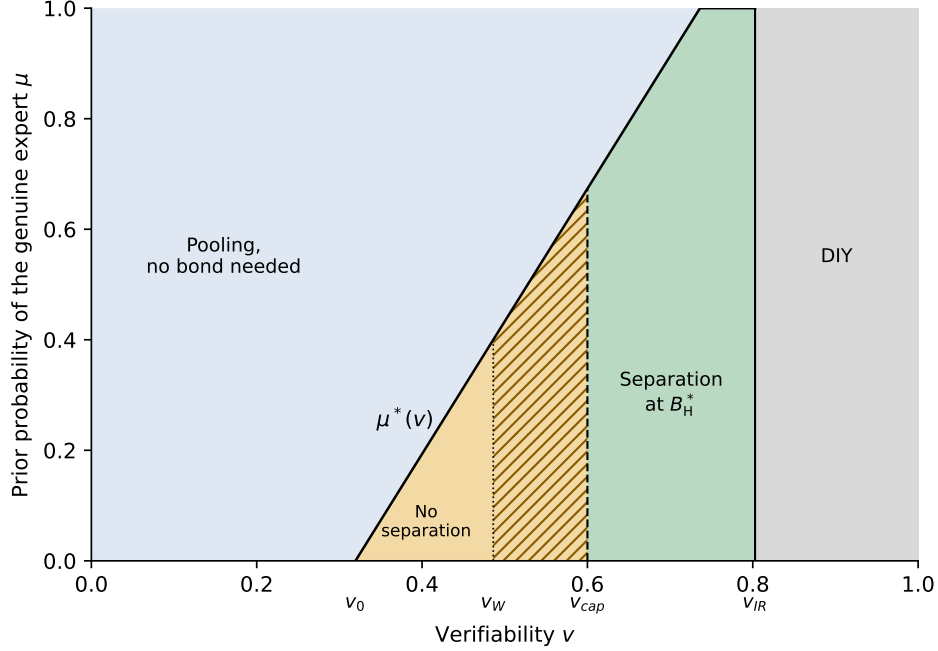


Figure 1: The (v, μ) map of Proposition 6 for the example of Appendix A.6 at $t = 1.2$. Separation at B_H^* occurs in the band $v_{\text{cap}} = 0.600 \leq v \leq v_{\text{IR}} = 0.803$, below $\mu^*(v)$. Above $\mu^*(v)$ the types pool without a bond; to the right of v_{IR} the principal does the task himself. Below $\mu^*(v)$ and to the left of the band there is no separation. In the hatched strip $[v_W, v_{\text{cap}}) = [0.486, 0.600)$, separation would occur if the expert could post any bond, and is lost to the capital constraint.

and high enough that the bond a court would have to enforce is one the expert can afford. Within the band, verifiability matters only through feasibility and not through the size of the principal's payoff (Proposition 3). The capital constraint of Assumption 4 matters when the separating bond becomes unaffordable before the wrapper can be engaged on its own, that is, when $v_{\text{cap}} > v_W$. In substance this is the judgment-proof problem (Shavell, 1986), applied to the credibility of a signal rather than to ex post compensation, which is why it interacts with the verification parameter v rather than standing alone.

Two implications for the AI context follow from the band's lower edge. Better imitation, a lower f_L , raises both v_W , because a known wrapper beats DIY at higher verification levels, and v_{cap} , because deterring a better wrapper requires a larger bond. More bonding capacity lowers v_{cap} , and so widens the band from below wherever capital is what sets the edge.

The paper takes the prior μ as given and asks where the market separates. Where it does not, the market cannot tell genuine experts from imitators. Because the price is fixed, a genuine expert earns no more than a wrapper in the pooling region, so it is not rewarded for being genuine. We conjecture that a market that cannot reward expertise will eventually lose it, since acquiring expertise brings no return. That would leave a market of DIY and wrappers, Akerlof's mechanism operating on the *investment* margin rather than on the quality of goods traded. The model does not derive this. It is a conjecture, and formalizing it is left for future work.

One assumption deserves a direct flag rather than a footnote. Assumption 5 selects the separating equilibrium wherever pooling without a bond is not enough, even where pooling at a positive bond also survives the intuitive criterion. There is a substantive reason beyond the refinement gap: pooling clears the principal’s participation constraint only in expectation, and a principal who cannot restore the underlying harm by a monetary transfer, or who is evaluated ex post rather than ex ante, has reason to reject it even when $\bar{U}$ is met on average. Section 6 returns to this.

6 Extensions

- **Costly bonds.** Real bonds carry costs roughly proportional to their face value, such as surety premiums, letter-of-credit fees, or the opportunity cost of collateral. Since the face value of B_H^* scales with $1/e$, weaker enforcement then makes separation more wasteful even when it is affordable, and the invariance of Proposition 3 no longer holds.
- **Endogenous verification.** Verifiability v is a task-level primitive here. Letting the principal invest in his own verification capacity opens a second under-investment problem, in the capacity to *check* alongside the capacity to *produce*.
- **Correlated failure and bond credibility.** The capacity ceiling $\bar{w}$ is exogenous here. A large firm that posts a large nominal bond it could not honor in a correlated failure is a distinct threat to the signal’s credibility.
- **Liability as one signal among several.** The analysis concerns a liability bond. Reputation, credentials, and institutional affiliation are alternative signals with verification requirements of their own, and AI erodes them through different channels. The paper does not compare them.
- **Risk and the ex-post cost of pooling.** The expected-value participation constraint suits the intermediary, whose bond-posting decision is naturally evaluated over repeated trials, but is a strong assumption for a one-shot principal, particularly where harm is not restorable by a transfer, as with a misdiagnosis rather than a mispriced audit. Pooling does not reveal to any given principal which type he engaged, so some fraction of the pool bears the full harm from a wrapper the bond never separated out. Replacing the expected-value participation constraint with a risk-averse or ex-post welfare criterion is left for future work.

7 Conclusion

Two mechanisms jointly determine whether a market for genuine expertise survives the arrival of cheap, competent AI execution: the principal’s ability to catch and redo a bad result himself, and a bond that lets genuine expertise be credibly signaled when he cannot. This paper shows that the bond separates genuine expertise from imitation only in an intermediate band of verifiability. Above the band, the principal does the task himself. Below it, a wrapper cannot be screened out, and a limit on the capital a genuine expert can post raises the band’s lower edge further. Better

imitation, which AI gives the wrapper, raises that edge; more bonding capacity lowers it wherever capital is what sets it. The model studies liability as the signal, takes verification as a task-level given, and fixes the price. Where separation fails, the market cannot tell experts from imitators, and we conjecture that it will not keep its experts.

AI Use Disclosure

Generative AI tools were used as a collaborative writing assistant to aid in the extensive drafting, structural redaction, and refinement of the manuscript text, including mathematical derivations. Specifically, Claude’s Opus 5 and Sonnet 5 models were used during manuscript preparation. These tools were used through an iterative process closely guided and prompted by the author. Outputs were iteratively reviewed, edited, and verified by the author. The author independently verified all references and claims, and is solely responsible for the content of the manuscript.

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A Proofs

A.1 Proof of Proposition 1

With the intermediary’s type known there is nothing to signal. At the posted price and with no bond, the delegated payoff is (2) at the known type with $B = 0$, and, by Assumption 3, it has no

retry term:

$$U_{\text{DEL}} = S - t - f_1 L, \quad f_1 \equiv 1 - p_1,$$

which does not depend on v . The DIY payoff (1) is affine in v :

$$U_{\text{DIY}}(v) = S - f_0 L - c_0 + v f_0 (L - \kappa).$$

Their difference is

$$U_{\text{DEL}} - U_{\text{DIY}}(v) = \Delta p \cdot L - \Delta c - v f_0 (L - \kappa), \quad \Delta p = f_0 - f_1, \quad \Delta c = t - c_0.$$

Given $L > \kappa$ the slope is strictly negative, so the difference changes sign exactly once, at v^* in (3). Below v^* , delegating at the posted price is strictly better than DIY. Above v^* , the principal walks away at $B = 0$. A known intermediary could then post a bond, which acts as an expected transfer to the principal, but the smallest bond that secures engagement leaves the principal exactly at $U_{\text{DIY}}(v)$. Delegation therefore cannot make him better off than DIY. (If $\kappa > L$ the slope reverses and so does the direction of the comparison; the case $L > \kappa$ is maintained throughout.)

A.2 Proof of Proposition 2

The marginal cost to type θ of posting bond B is $\partial(f_\theta e B)/\partial B = f_\theta e$, independent of B . Costs are linear, so single crossing holds trivially: H's cost lies strictly below L's at every $B > 0$.

Consider a wrapper that is identified as such. The principal engages it at bond B if and only if $S - t - f_L L + f_L e B \geq \bar{U}$. At the smallest such bond, the wrapper earns

$$t - f_L e B = t - (\bar{U} - S + t + f_L L) = S - f_L L - \bar{U},$$

and any larger bond earns it less. Under $S - f_L L \leq \bar{U}$ an identified wrapper therefore cannot earn a positive payoff, and its payoff when separated is 0. Mimicking a bond B that identifies H is deterred if and only if

$$t - f_L e B \leq 0 \iff B \geq \frac{t}{f_L e}.$$

H wants the smallest bond satisfying this, since bonding is costly, so $B_H^* = t/(f_L e)$. Its payoff there is $t - f_H e B_H^* = t(1 - f_H/f_L) > 0$. Engagement at B_H^* additionally requires $U(H, B_H^*) \geq \bar{U}$. Separating equilibria at bonds above B_H^* fail the intuitive criterion: a deviation to a bond between B_H^* and the equilibrium bond is equilibrium-dominated for L but not for H, so the principal's belief must place it on H, and H gains by deviating. Off-path bonds below B_H^* can be attributed to L, since L would gain from them if engaged. Hence, as in Riley (1979), B_H^* is the unique separating bond that survives the intuitive criterion.

A.3 Proof of Proposition 4

When both types pool at $B = 0$, the principal's posterior equals the prior, and his payoff is

$$U_{\text{pool}}(\mu) = S - t - [\mu f_H + (1 - \mu) f_L]L,$$

which is increasing in μ since $f_H < f_L$. Setting $U_{\text{pool}}(\mu) = \bar{U}$ and solving for μ gives (5), and $\mu \geq \mu^*$ satisfies the constraint.

A.4 Proof of Proposition 5

Pooling threshold. Substituting $\bar{U} = U_{\text{DIY}}(v)$ into (5),

$$S - t - U_{\text{DIY}}(v) = (c_0 - t) + (1 - v)f_0L + vf_0\kappa,$$

so

$$\mu^*(v) = \frac{f_L - \frac{(c_0 - t) + (1 - v)f_0L + vf_0\kappa}{L}}{f_L - f_H} = \frac{f_L - f_0 + \frac{t - c_0}{L} + v \frac{f_0(L - \kappa)}{L}}{f_L - f_H} = \frac{vf_0(L - \kappa) - \Delta + t}{L(f_L - f_H)}.$$

This is affine in v with slope $f_0(L - \kappa)/[L(f_L - f_H)] > 0$ when $L > \kappa$.

Participation boundary. By Proposition 3, $U(H, B_H^*) = S - t - f_H L + f_H t / f_L$ does not depend on v , while $U_{\text{DIY}}(v)$ is increasing. They cross once, at v_{IR} in (7); the principal prefers an identified H at B_H^* to DIY if and only if $v \leq v_{\text{IR}}$. Identity (8) follows by evaluating (3) at $f_1 = f_H$ and comparing numerators.

Ordering. Let v_0 solve $\mu^*(v_0) = 0$, so $v_0 = (\Delta - t)/[f_0(L - \kappa)]$. Then

$$v_W - v_0 = \frac{t}{f_0(L - \kappa)} > 0, \quad v_{\text{IR}} - v_W = \frac{(f_L - f_H)(L - t/f_L)}{f_0(L - \kappa)} > 0,$$

the second since $f_L L > t$. Hence $v_0 < v_W < v_{\text{IR}}$, and $\mu^*(v) > 0$ throughout the band. Moreover, $\mu^*(v_{\text{IR}}) = 1 + (t/L)[1/(f_L - f_H) - 1/f_L] > 1$, so the pooling region does not intersect $v \geq v_{\text{IR}}$.

(a) *Pooling without a bond.* For $\mu \geq \mu^*(v)$, pooling at $B = 0$ satisfies the principal's participation constraint (Proposition 4). No type gains by deviating to a positive bond, since it is already engaged at t . This is the outcome selected by Assumption 5(i).

(b) *Separating band.* For $\mu < \mu^*(v)$, pooling at $B = 0$ fails. If $v_W \leq v \leq v_{\text{IR}}$, the scope condition of Proposition 2 holds and $U(H, B_H^*) \geq U_{\text{DIY}}(v)$, so H separates at B_H^* . This is the outcome selected by Assumption 5(ii).

(c) *No separation.* For $\mu < \mu^*(v)$ and $v < v_W$, an identified wrapper can obtain engagement and earn $S - f_L L - U_{\text{DIY}}(v) > 0$. No separating equilibrium in which the wrapper earns zero exists, so the deterrence argument of Proposition 2 fails. By Assumption 5, we do not characterize the outcome.

(d) *DIY.* Let $v > v_{\text{IR}}$. Any bond $B \leq B_H^*$ that both types might post yields the principal

$S - t - \bar{f}(L - eB)$, where $\bar{f} \equiv \mu f_H + (1 - \mu)f_L$, which is at most $S - t - f_H(L - eB_H^*) = U(H, B_H^*) < U_{\text{DIY}}(v)$, since $eB \leq t/f_L < L$. A wrapper will not post a bond above B_H^* . An identified H can still obtain engagement by posting a bond above B_H^* , which acts as an expected transfer to the principal, but the smallest such bond leaves him exactly at $U_{\text{DIY}}(v)$. Delegation therefore cannot make the principal better off than DIY.

A.5 Proof of Proposition 6

Separation at B_H^* requires $B_H^* \leq \bar{w}$. With $e(v) = v$, $B_H^* = t/(f_L v) \leq \bar{w}$ if and only if $v \geq v_{\text{cap}} = t/(f_L \bar{w})$. Combined with Proposition 5, H separates at B_H^* exactly when $\mu < \mu^*(v)$ and $\max\{v_W, v_{\text{cap}}\} \leq v \leq v_{\text{IR}}$. Since $v_W < v_{\text{IR}}$, this interval is empty if and only if $v_{\text{cap}} > v_{\text{IR}}$. For $\mu < \mu^*(v)$ and $v < \max\{v_W, v_{\text{cap}}\}$, there is no separating equilibrium: for $v < v_W$ by Proposition 5, and for $v_W \leq v < v_{\text{cap}}$ because deterring the wrapper requires a bond of at least $B_H^* > \bar{w}$. The pooling and DIY regions are unchanged, since neither $\mu^*(v)$ nor v_{IR} involves $\bar{w}$.

A.6 Numerical example

Let $f_H = 0.2$, $f_L = 0.5$, $f_0 = 0.8$, $L = 10$, $\kappa = 1$, $c_0 = 0.5$, $\bar{w} = 4$, and $S = 20$. Then $\Delta = 3.5$ and $v_W = 0.486$.

Table 1: Separating band in the numerical example

t	v_0	v_W	v_{cap}	v_{IR}	Separating band
0.50	0.417	0.486	0.250	0.861	[0.486, 0.861]
0.80	0.375	0.486	0.400	0.836	[0.486, 0.836]
1.20	0.319	0.486	0.600	0.803	[0.600, 0.803]
1.50	0.278	0.486	0.750	0.778	[0.750, 0.778]

At $t = 0.5$ and $t = 0.8$, $v_{\text{cap}} \leq v_W$: capital is slack and the band is the baseline $[v_W, v_{\text{IR}}]$. At $t = 1.2$ and $t = 1.5$, $v_{\text{cap}} > v_W$ and capital cuts the band from below; at $t = 1.5$ only a sliver of width 0.028 remains. Figure 1 plots the case $t = 1.2$.

A.7 Robustness to the enforcement technology

Take a general affine specification, $e(v) = e_0 + (1 - e_0)v$. Neither v_W nor v_{IR} involves e : the first compares a known wrapper with DIY, and the second rests on the e -invariance of Proposition 3. The pooling threshold $\mu^*(v)$ does not involve e either, since it concerns pooling at $B = 0$. Only the capacity threshold changes, to

$$v_{\text{cap}} = \frac{\hat{e} - e_0}{1 - e_0}, \quad \hat{e} = \frac{t}{f_L \bar{w}},$$

so the separating band is still $\max\{v_W, v_{\text{cap}}\} \leq v \leq v_{\text{IR}}$ below $\mu^*(v)$.